

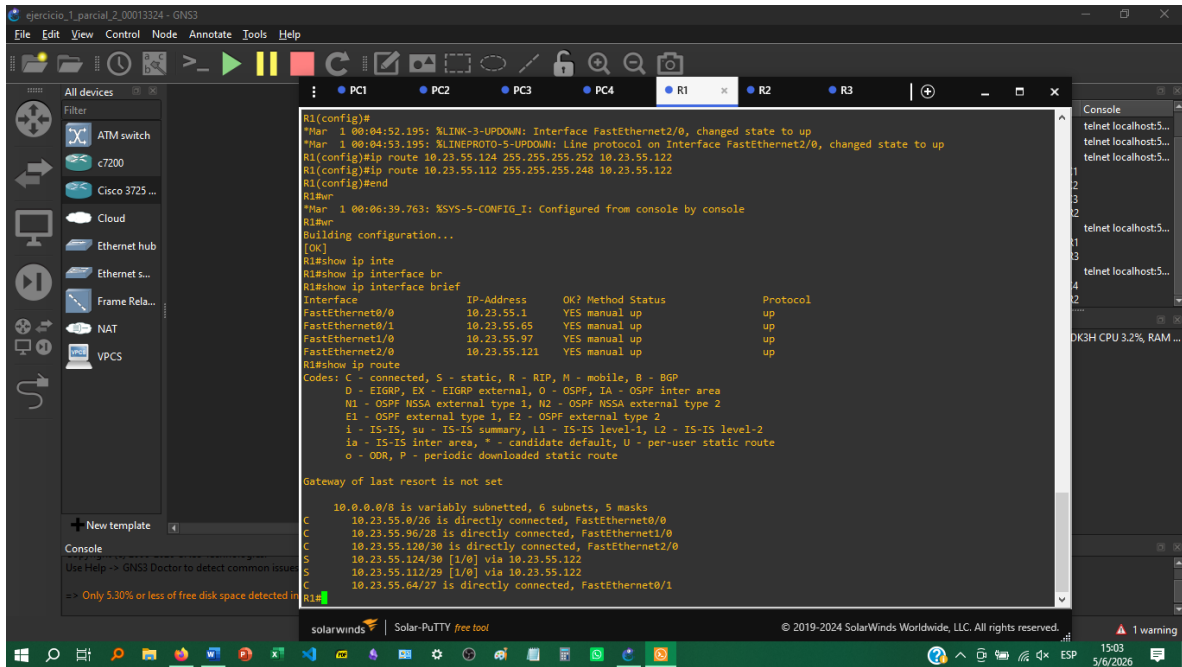
Capturas de pantalla Parcial 2

Realizado por: Carlos Humberto Cruz Parada (00013324)

Ejercicio 1:

Rutas:

- R1:



```
ejercicio_1_parcial_2_00013324 - GNS3
File Edit View Control Node Annotate Tools Help

All devices
Filter
ATM switch
c7200
Cisco 3725 ...
Cloud
Ethernet hub
Ethernet s...
Frame Rela...
NAT
VPCS

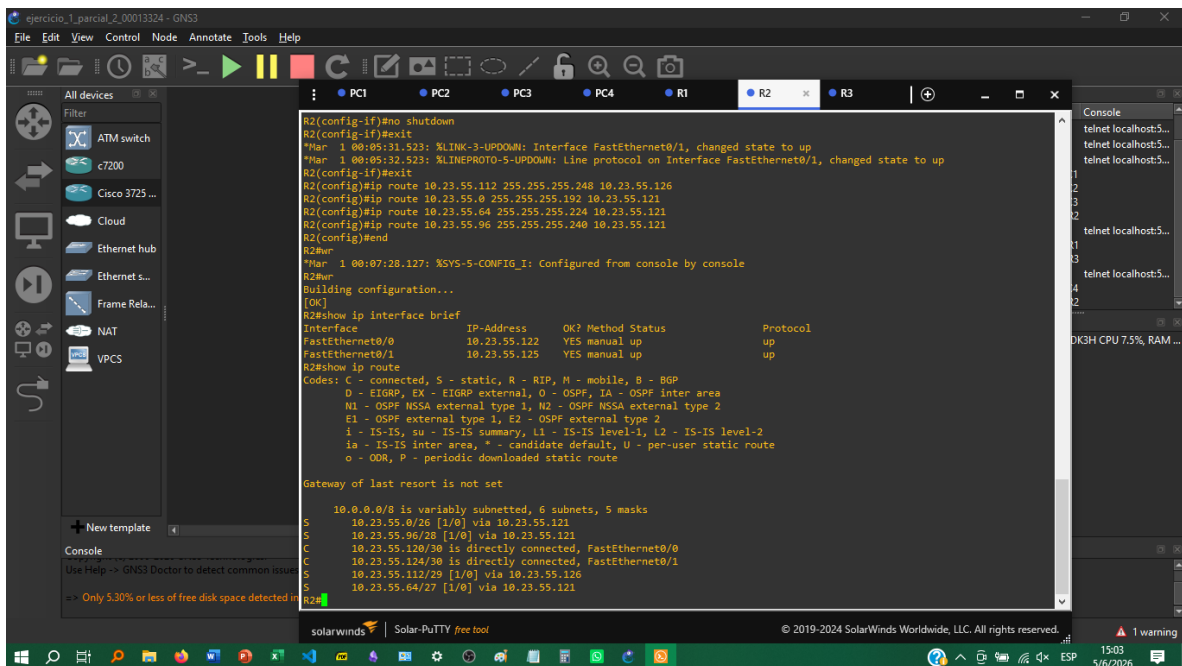
+ New template
Console
Use Help -> GNS3 Doctor to detect common issues
=> Only 5.30% or less of free disk space detected

R1
R1(config)#
*Mar 1 00:04:52.195: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Mar 1 00:04:53.195: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R1(config)#ip route 10.23.55.124 255.255.255.252 10.23.55.122
R1(config)#ip route 10.23.55.112 255.255.255.248 10.23.55.122
R1(config)#end
R1#
*Mar 1 00:06:39.763: %SYS-5-CONFIG_I: Configured from console by console
R1#
Building configuration...
[OK]
R1#show ip inte
R1#show ip interface br
R1#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    10.23.55.1      YES manual up          up
FastEthernet0/1    10.23.55.69     YES manual up          up
FastEthernet1/0    10.23.55.97     YES manual up          up
FastEthernet2/0    10.23.55.121    YES manual up          up
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, Ek - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C       10.23.55.0/26 is directly connected, FastEthernet0/0
C       10.23.55.96/28 is directly connected, FastEthernet1/0
C       10.23.55.120/30 is directly connected, FastEthernet2/0
S       10.23.55.124/30 [1/0] via 10.23.55.122
S       10.23.55.112/29 [1/0] via 10.23.55.122
C       10.23.55.64/27 is directly connected, FastEthernet0/1
R1#
```

- R2:



```
ejercicio_1_parcial_2_00013324 - GNS3
File Edit View Control Node Annotate Tools Help

All devices
Filter
ATM switch
c7200
Cisco 3725 ...
Cloud
Ethernet hub
Ethernet s...
Frame Rela...
NAT
VPCS

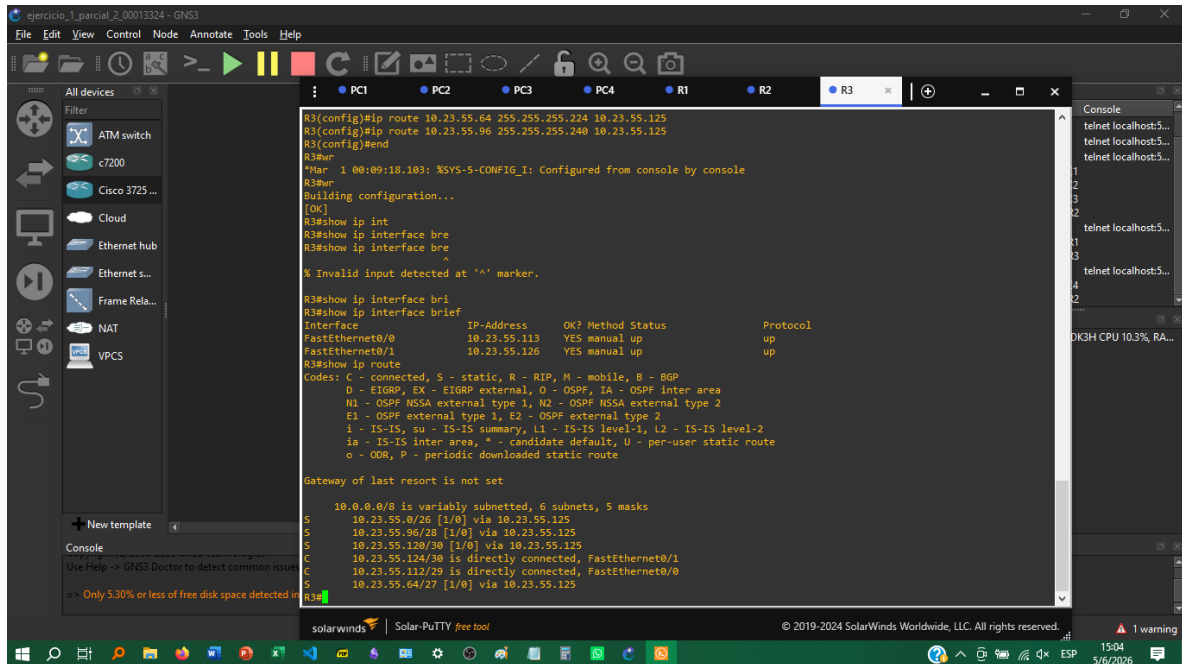
+ New template
Console
Use Help -> GNS3 Doctor to detect common issues
=> Only 5.30% or less of free disk space detected

R2
R2(config-if)#no shutdown
R2(config-if)#exit
*Mar 1 00:05:31.523: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:05:32.523: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config-if)#exit
R2(config)#ip route 10.23.55.112 255.255.255.248 10.23.55.126
R2(config)#ip route 10.23.55.0 255.255.255.192 10.23.55.121
R2(config)#ip route 10.23.55.64 255.255.255.224 10.23.55.121
R2(config)#ip route 10.23.55.96 255.255.255.240 10.23.55.121
R2(config)#end
R2#
*Mar 1 00:07:28.127: %SYS-5-CONFIG_I: Configured from console by console
R2#
Building configuration...
[OK]
R2#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    10.23.55.122    YES manual up          up
FastEthernet0/1    10.23.55.125    YES manual up          up
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, Ek - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

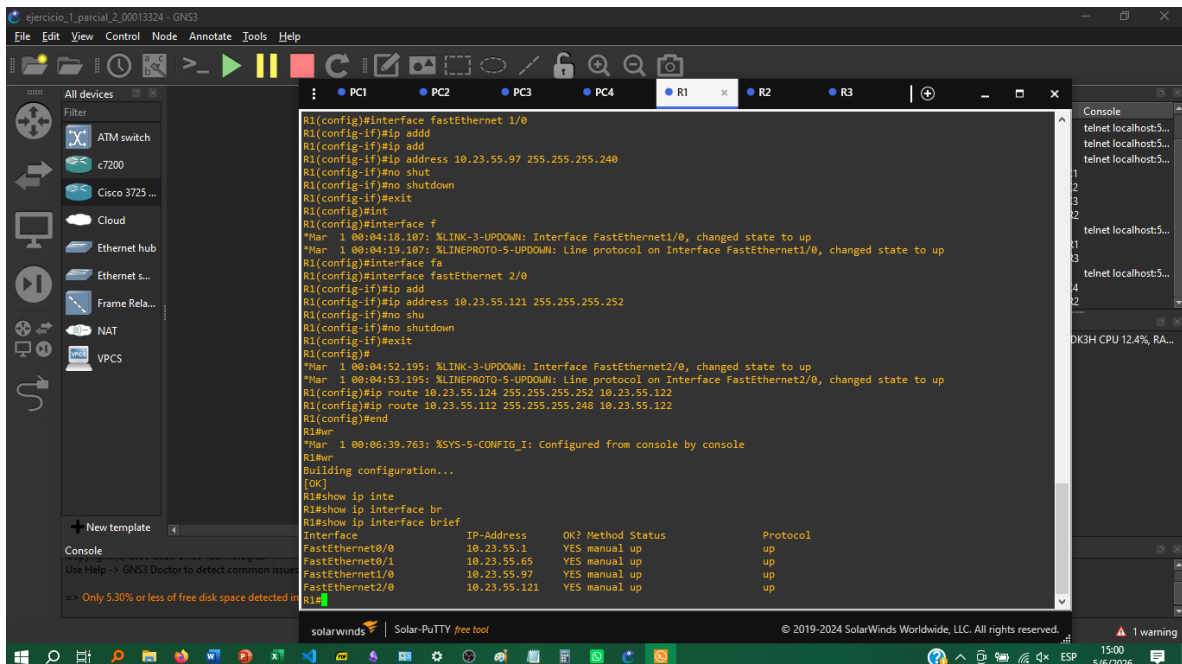
10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S       10.23.55.0/26 [1/0] via 10.23.55.121
S       10.23.55.96/28 [1/0] via 10.23.55.121
C       10.23.55.120/30 is directly connected, FastEthernet0/0
C       10.23.55.124/30 is directly connected, FastEthernet0/1
S       10.23.55.112/29 [1/0] via 10.23.55.126
S       10.23.55.64/27 [1/0] via 10.23.55.121
R2#
```

- R3:

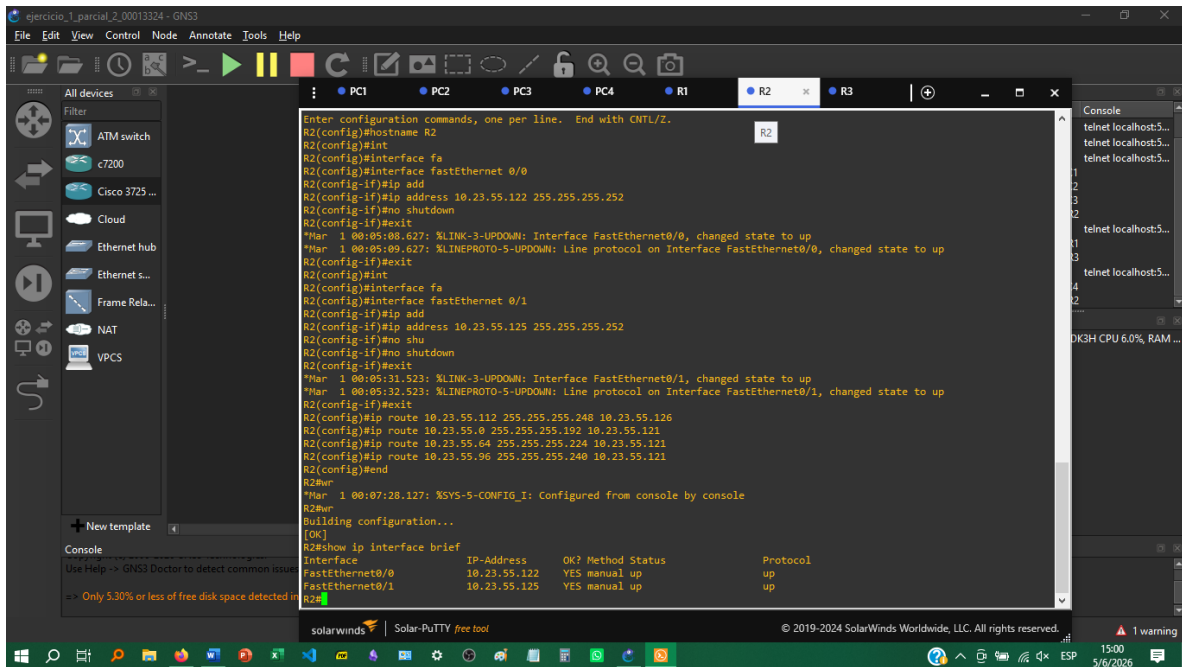


Interfaces:

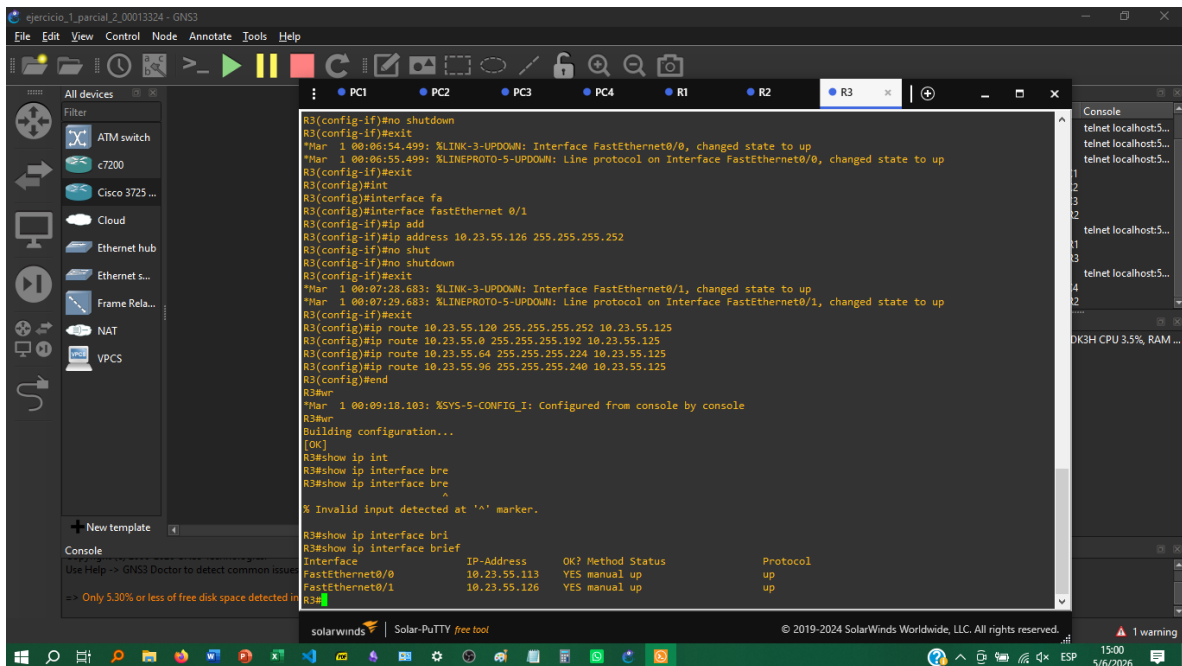
- R1:



- R2:

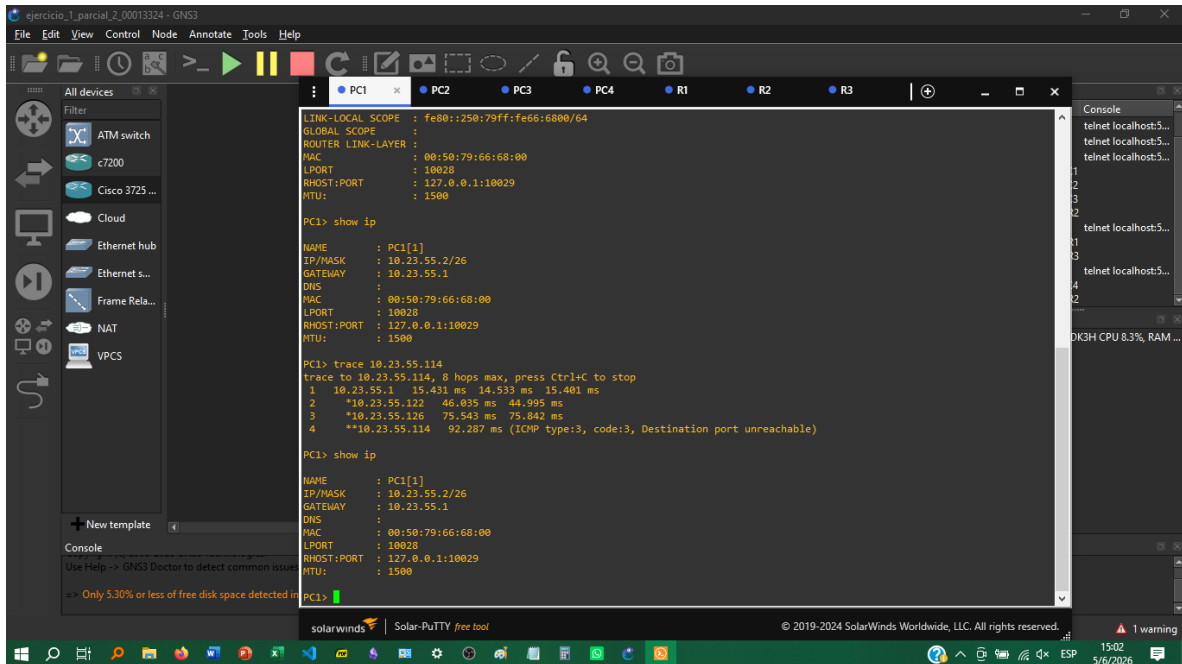


- R3:

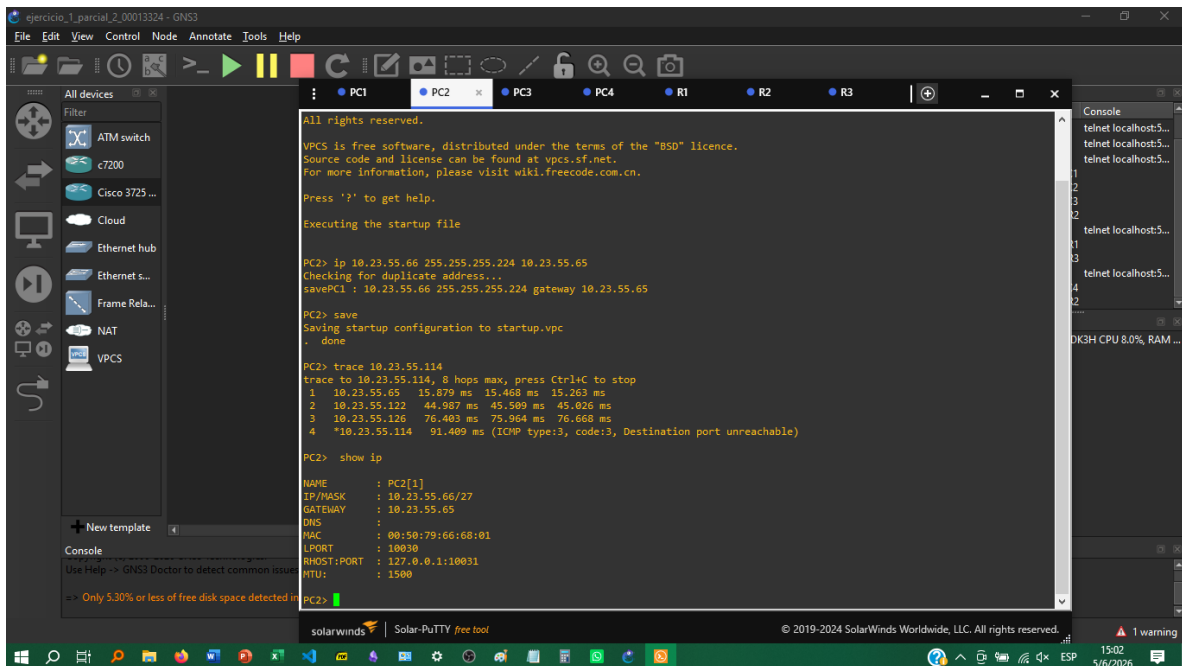


IP de PC:

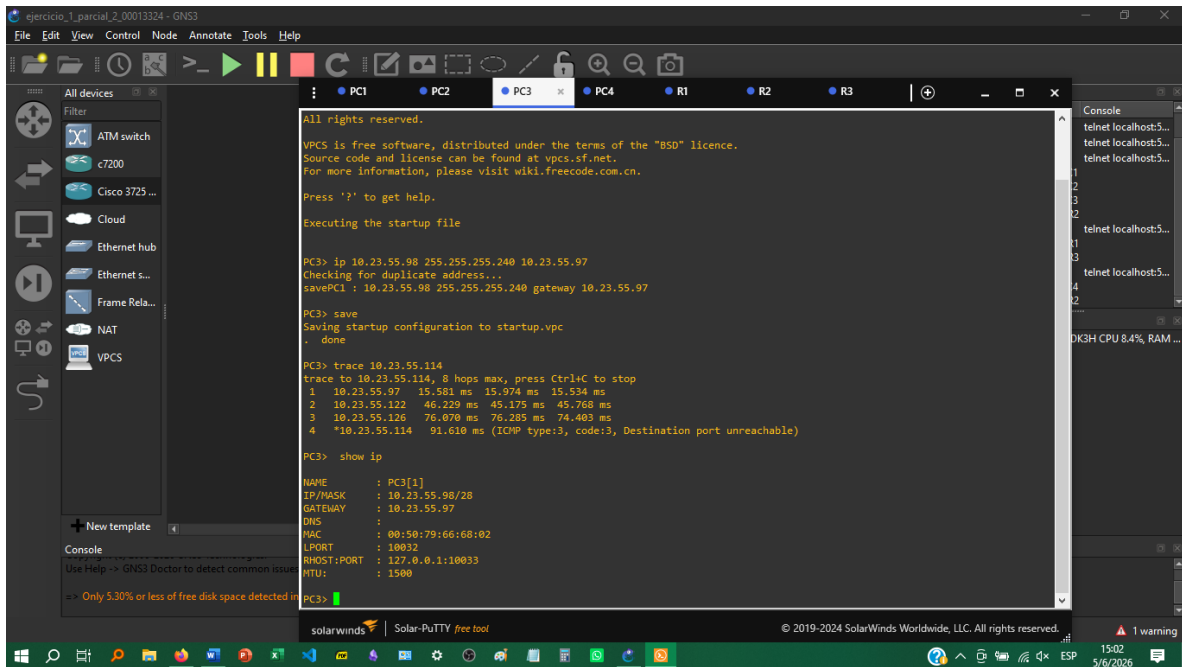
- PC 1:



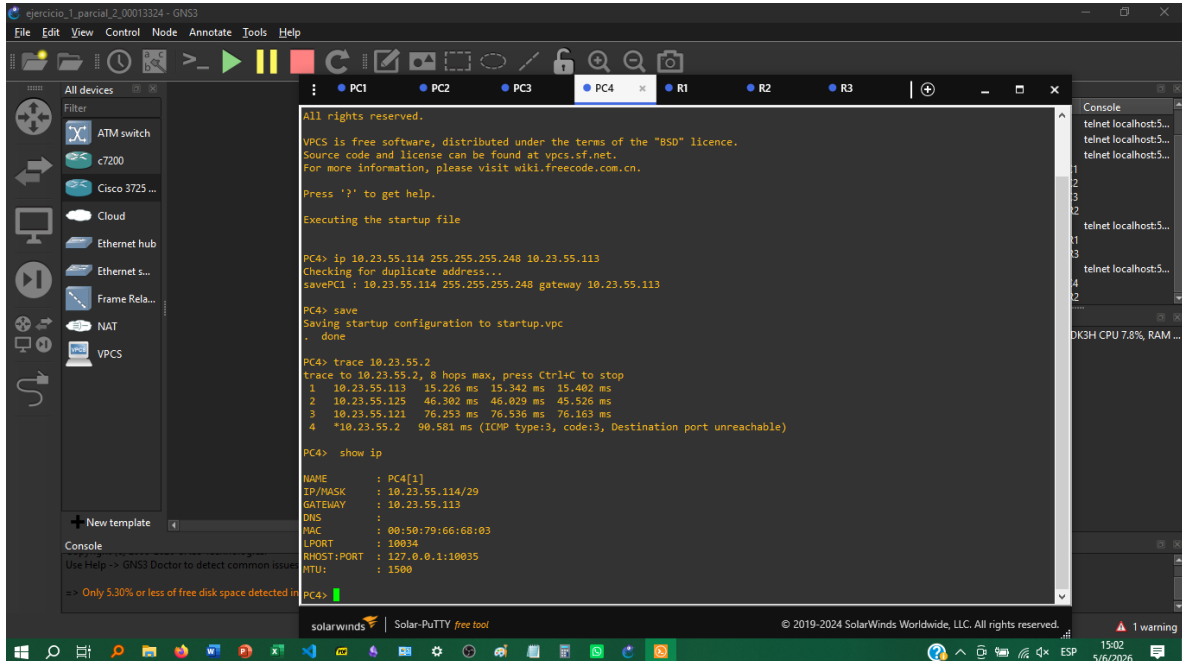
- PC 2:



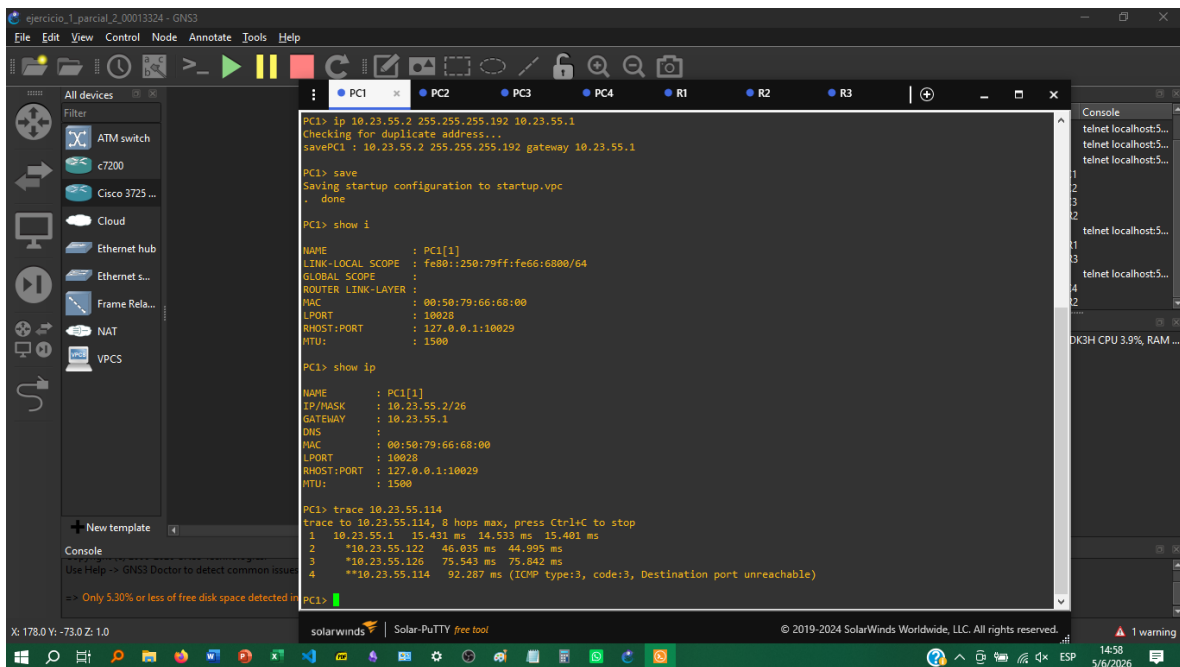
- PC 3:



- PC 4:



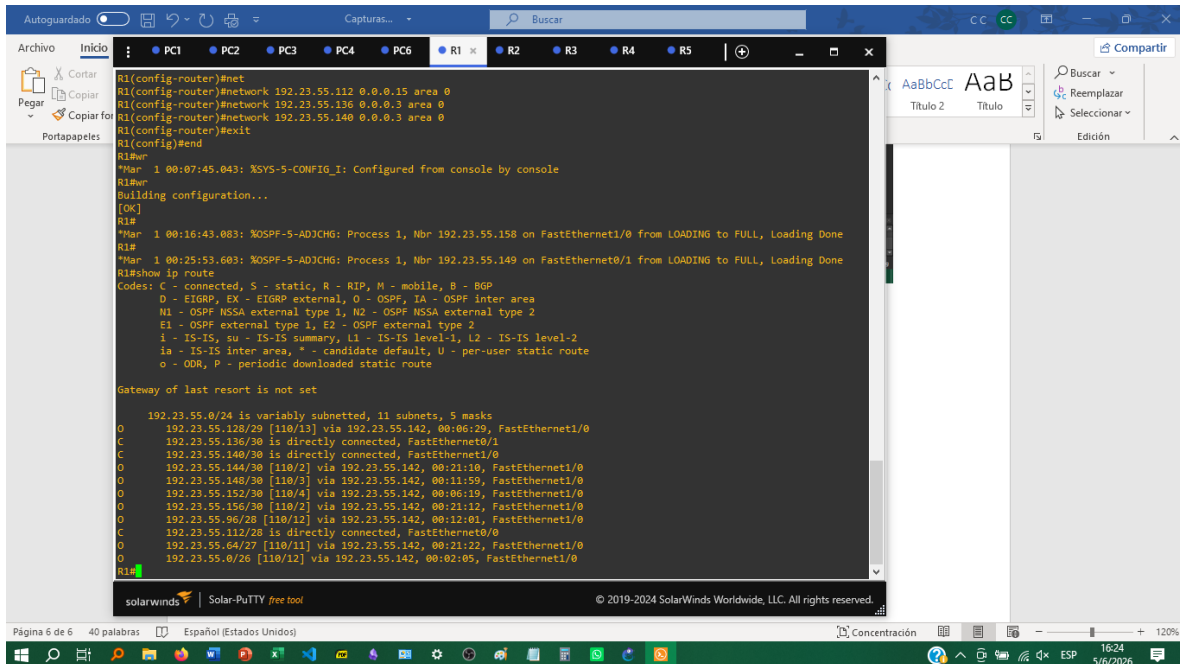
Trace de IP:



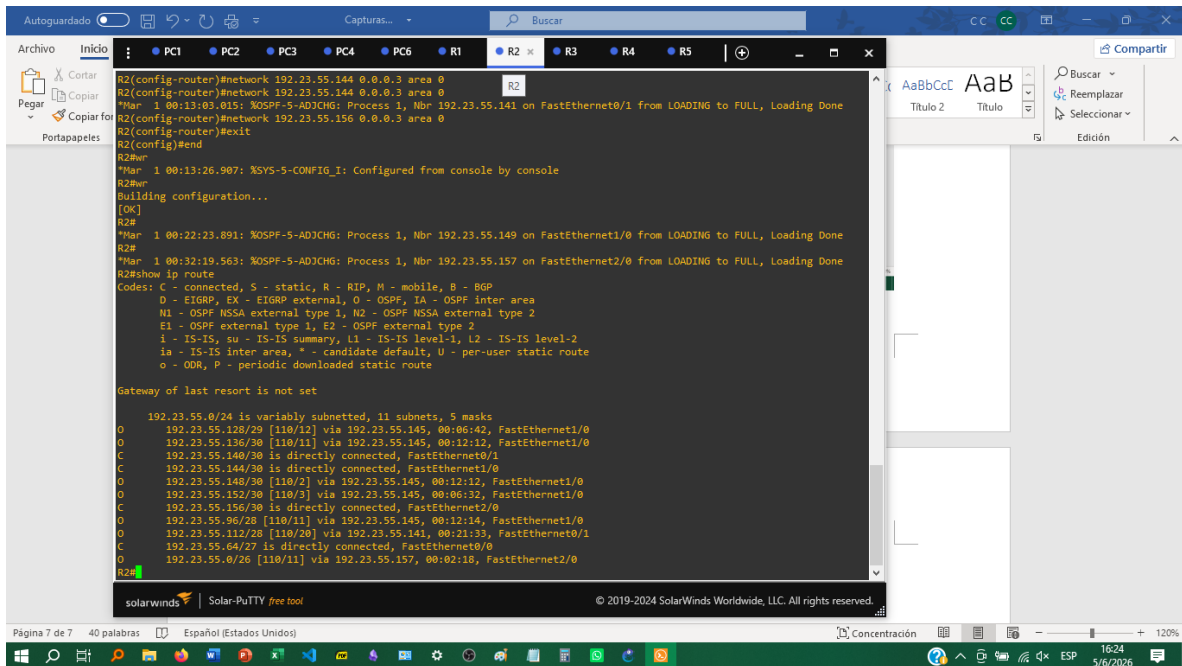
Ejercicio 2:

Rutas:

R1:



R2:



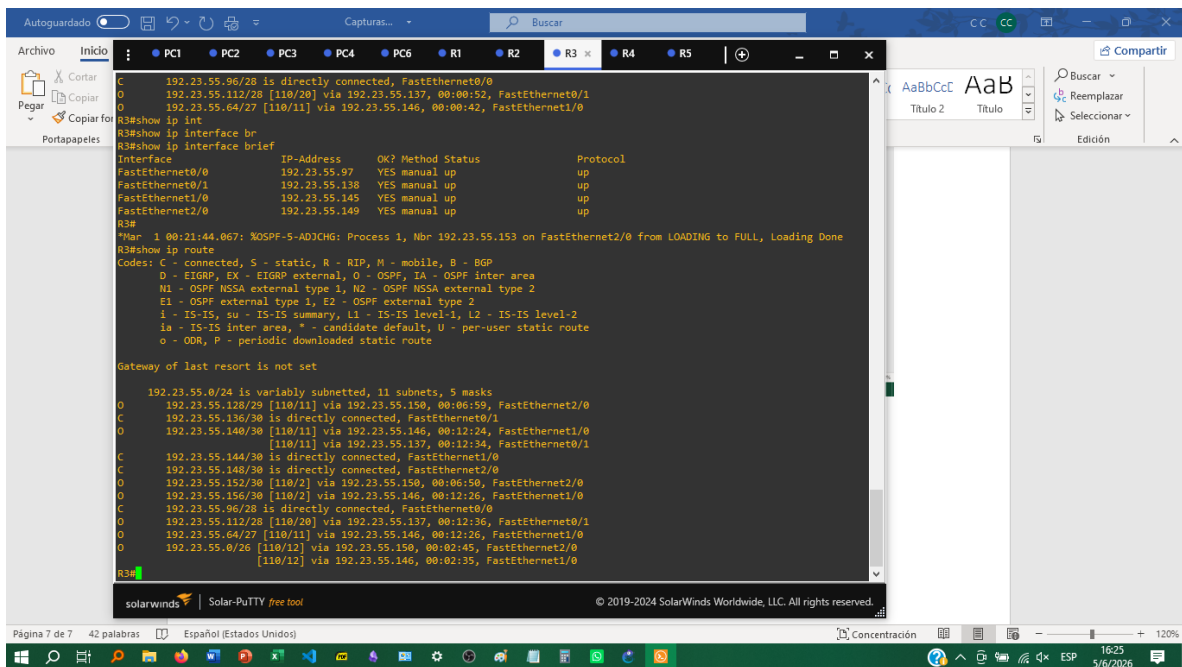
The screenshot shows a SolarWinds Solar-PuTTY terminal window with the title bar "Autoguardado" and "Capturas...". The terminal is connected to a device named "R2". The user has entered the command "R2(config-router)#network 192.23.55.144 0.0.0.3 area 0", and the output shows the configuration being applied. The user then enters "R2(config-router)#exit" and "R2(config)#end". The terminal displays the configuration summary for R2, including OSPF settings and a list of connected interfaces. The status bar at the bottom indicates "Página 7 de 7", "40 palabras", and "Español (Estados Unidos)".

```
R2(config-router)#network 192.23.55.144 0.0.0.3 area 0
R2(config-router)#network 192.23.55.144 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#end
R2#
*Mar 1 00:13:26.907: %SYS-5-CONFIG_I: Configured from console by console
R2#
Building configuration...
R2#
*Mar 1 00:22:23.891: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.55.149 on FastEthernet1/0 from LOADING to FULL, Loading Done
R2#
*Mar 1 00:32:19.563: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.55.157 on FastEthernet2/0 from LOADING to FULL, Loading Done
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.55.0/24 is variably subnetted, 11 subnets, 5 masks
O       192.23.55.128/29 [110/12] via 192.23.55.145, 00:06:42, FastEthernet1/0
O       192.23.55.136/30 [110/11] via 192.23.55.145, 00:12:12, FastEthernet1/0
C       192.23.55.140/30 is directly connected, FastEthernet0/1
C       192.23.55.144/30 is directly connected, FastEthernet1/0
O       192.23.55.148/30 [110/2] via 192.23.55.145, 00:12:12, FastEthernet1/0
O       192.23.55.152/30 [110/3] via 192.23.55.145, 00:06:32, FastEthernet1/0
C       192.23.55.156/30 is directly connected, FastEthernet2/0
O       192.23.55.96/28 [110/11] via 192.23.55.145, 00:12:14, FastEthernet1/0
O       192.23.55.112/28 [110/20] via 192.23.55.141, 00:21:33, FastEthernet0/1
C       192.23.55.64/27 is directly connected, FastEthernet0/0
O       192.23.55.0/26 [110/11] via 192.23.55.137, 00:02:10, FastEthernet2/0
R2#
```

R3:



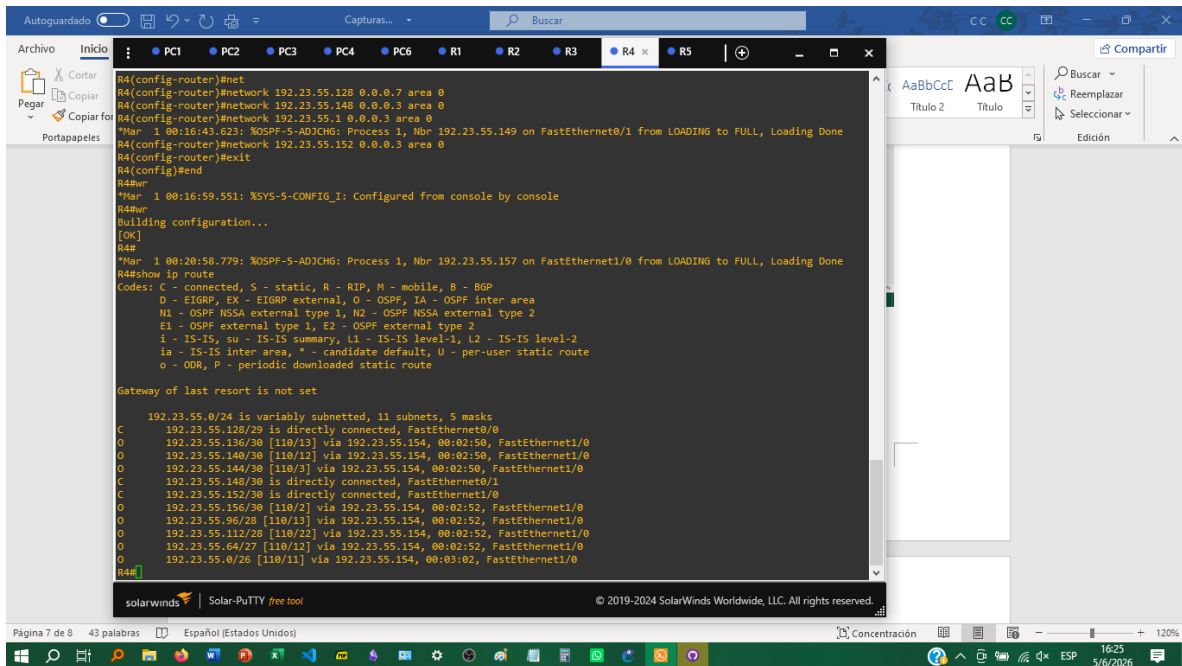
The screenshot shows a SolarWinds Solar-PuTTY terminal window with the title bar "Autoguardado" and "Capturas...". The terminal is connected to a device named "R3". The user has entered the command "R3#show ip int", and the output shows the configuration for the interfaces. The user then enters "R3#show ip interface brief", and the output shows a table of interfaces and their status. The user then enters "R3#show ip route", and the output shows the configuration summary for R3, including OSPF settings and a list of connected interfaces. The status bar at the bottom indicates "Página 7 de 7", "42 palabras", and "Español (Estados Unidos)".

```
R3#show ip int
R3#show ip interface brief
R3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.23.55.97    YES manual up          up
FastEthernet0/1 192.23.55.138  YES manual up          up
FastEthernet1/0 192.23.55.145  YES manual up          up
FastEthernet2/0 192.23.55.149  YES manual up          up
R3#
*Mar 1 00:21:44.067: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.55.153 on FastEthernet2/0 from LOADING to FULL, Loading Done
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.55.0/24 is variably subnetted, 11 subnets, 5 masks
O       192.23.55.128/29 [110/11] via 192.23.55.150, 00:06:59, FastEthernet2/0
O       192.23.55.136/30 is directly connected, FastEthernet0/1
O       192.23.55.140/30 [110/11] via 192.23.55.146, 00:12:24, FastEthernet1/0
O       192.23.55.148/30 [110/11] via 192.23.55.137, 00:12:34, FastEthernet0/1
C       192.23.55.144/30 is directly connected, FastEthernet1/0
O       192.23.55.148/30 is directly connected, FastEthernet2/0
O       192.23.55.152/30 [110/2] via 192.23.55.150, 00:06:50, FastEthernet2/0
O       192.23.55.156/30 [110/2] via 192.23.55.146, 00:12:26, FastEthernet1/0
C       192.23.55.96/28 is directly connected, FastEthernet0/0
O       192.23.55.112/28 [110/20] via 192.23.55.137, 00:12:36, FastEthernet0/1
O       192.23.55.64/27 [110/11] via 192.23.55.146, 00:12:26, FastEthernet1/0
O       192.23.55.0/26 [110/12] via 192.23.55.150, 00:02:45, FastEthernet2/0
O       192.23.55.0/26 [110/12] via 192.23.55.146, 00:02:35, FastEthernet1/0
R3#
```

R4:



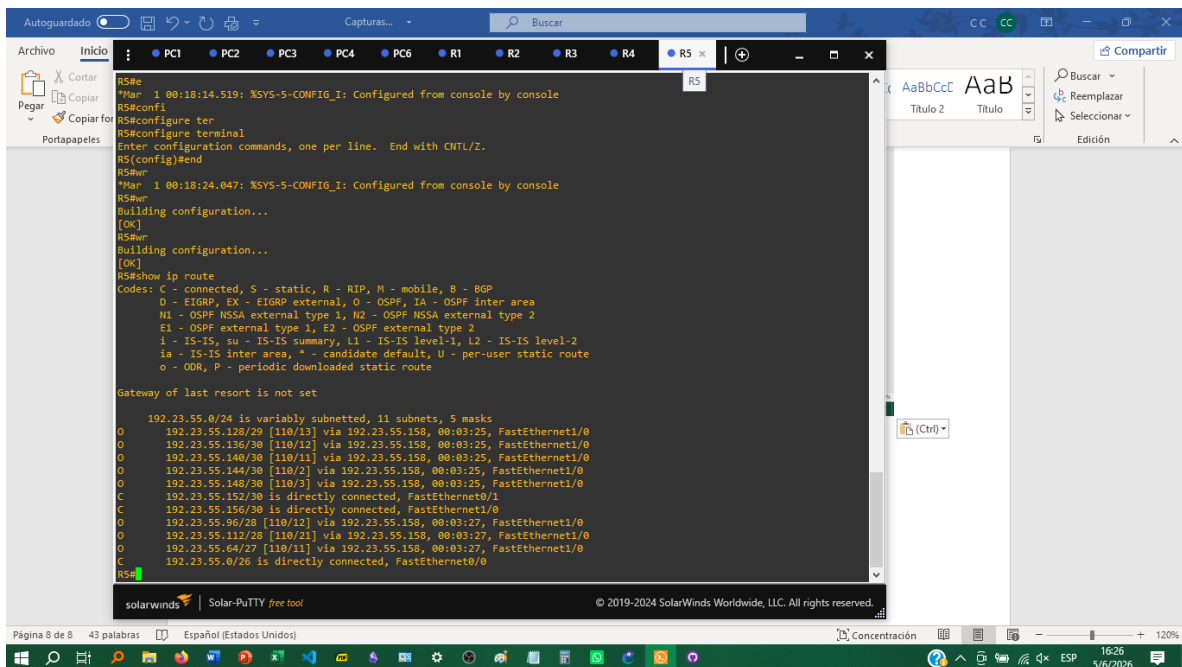
The screenshot shows a SolarWinds Solar-PuTTY terminal window with the title bar "Capturas...". The window is displaying the configuration for router R4. The configuration includes setting the hostname to R4, configuring three interfaces (FastEthernet0/0, FastEthernet0/1, and FastEthernet0/26) with IP addresses and subnets, and enabling OSPF. The terminal output shows the configuration being applied and the router's status.

```
R4(config-router)#net
R4(config-router)#network 192.23.55.128 0.0.0.7 area 0
R4(config-router)#network 192.23.55.148 0.0.0.3 area 0
R4(config-router)#network 192.23.55.1 0.0.0.3 area 0
*Mar 1 00:16:43.623: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.55.149 on FastEthernet0/1 from LOADING to FULL, Loading Done
R4(config-router)#exit
R4(config)#end
R4#
*Mar 1 00:16:59.551: %SYS-5-CONFIG_I: Configured from console by console
R4#
Building configuration...
[OK]
R4#
*Mar 1 00:20:58.779: %OSPF-5-ADJCHG: Process 1, Nbr 192.23.55.157 on FastEthernet1/0 from LOADING to FULL, Loading Done
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.55.0/24 is variably subnetted, 11 subnets, 5 masks
C    192.23.55.128/29 is directly connected, FastEthernet0/0
O    192.23.55.136/30 [110/13] via 192.23.55.154, 00:02:50, FastEthernet1/0
O    192.23.55.140/30 [110/12] via 192.23.55.154, 00:02:50, FastEthernet1/0
O    192.23.55.144/30 [110/3] via 192.23.55.154, 00:02:50, FastEthernet1/0
O    192.23.55.148/30 is directly connected, FastEthernet0/1
C    192.23.55.152/30 is directly connected, FastEthernet1/0
O    192.23.55.156/30 [110/2] via 192.23.55.154, 00:02:52, FastEthernet1/0
O    192.23.55.96/28 [110/13] via 192.23.55.154, 00:02:52, FastEthernet1/0
O    192.23.55.112/28 [110/22] via 192.23.55.154, 00:02:52, FastEthernet1/0
O    192.23.55.64/27 [110/12] via 192.23.55.154, 00:02:52, FastEthernet1/0
O    192.23.55.0/26 [110/11] via 192.23.55.154, 00:03:02, FastEthernet1/0
R4#
```

R5:



The screenshot shows a SolarWinds Solar-PuTTY terminal window with the title bar "Capturas...". The window is displaying the configuration for router R5. The configuration includes setting the hostname to R5, configuring three interfaces (FastEthernet0/0, FastEthernet0/1, and FastEthernet0/26) with IP addresses and subnets, and enabling OSPF. The terminal output shows the configuration being applied and the router's status.

```
R5#
*Mar 1 00:18:14.519: %SYS-5-CONFIG_I: Configured from console by console
R5#confi
R5#configure ter
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#end
R5#
*Mar 1 00:18:24.047: %SYS-5-CONFIG_I: Configured from console by console
R5#
Building configuration...
[OK]
R5#
Building configuration...
[OK]
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.23.55.0/24 is variably subnetted, 11 subnets, 5 masks
O    192.23.55.128/29 [110/13] via 192.23.55.158, 00:03:25, FastEthernet1/0
O    192.23.55.136/30 [110/12] via 192.23.55.158, 00:03:25, FastEthernet1/0
O    192.23.55.140/30 [110/11] via 192.23.55.158, 00:03:25, FastEthernet1/0
O    192.23.55.144/30 [110/2] via 192.23.55.158, 00:03:25, FastEthernet1/0
O    192.23.55.148/30 [110/3] via 192.23.55.158, 00:03:25, FastEthernet1/0
C    192.23.55.152/30 is directly connected, FastEthernet0/1
C    192.23.55.156/30 is directly connected, FastEthernet1/0
O    192.23.55.96/28 [110/12] via 192.23.55.158, 00:03:27, FastEthernet1/0
O    192.23.55.112/28 [110/21] via 192.23.55.158, 00:03:27, FastEthernet1/0
O    192.23.55.64/27 [110/11] via 192.23.55.158, 00:03:27, FastEthernet1/0
C    192.23.55.0/26 is directly connected, FastEthernet0/0
R5#
```


Procesos: